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Mechanochemical synthesis of spinel cobalt ferrite nanopowders: Mathematical modeling and experiment

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ABSTRACT

The mechanochemical synthesis of a three-component powder mixture in a high-energy mill has been experimentally and theoretically studied using a macroscopic mathematical model. The modes of mechanochemical synthesis of nanoproducts with the required characteristics were determined. The discrete operating mode of the mill automatically ensures the development of the chemical process under "soft", almost isothermal conditions, rather than in an explosive mode. The addition of inert sodium chloride optimizes the mechanochemical synthesis of nanoparticles. An analytical formula linking the degree of mechanochemical conversion with the time of mechanical treatment and the concentration of the inert component in the starting mixture was derived. Theoretical calculations were analytically compared with experimental data on the mechanochemical synthesis of spinel cobalt ferrite nanopowders.

1. Introduction

Mechanochemical synthesis is a chemical process occurring during the mechanical treatment of a reacting mixture. Mechanical treatment leads to intense grinding and plastic deformation of reactants, acceleration of mass transfer due to diffusion and viscous flow of the substance, which contributes to the formation of composites and significantly increases the rate of chemical transformations [1–12].

Mechanical treatment can be used to control the chemical process and decrease the size of the particles, up to nanometers [13–16].

T. Tsuzuki et al. [14] reported a detailed review of the mechanochemical synthesis of nanoparticles, including metal systems, sulfides, carbonates and oxides, and concluded that mechanically synthesized particles had an average size of about 4 nm. In this case, the particles had low agglomeration, homogeneous structure, and morphology.

Mathematical modeling is widely used in mechanochemical synthesis. It is assumed that the mechanical energy transferred to the substance is converted into other forms of energy and the degree of mechanochemical conversion depends on the mechanical energy transferred to the substance being milled [17]. F. K. Urakaev et al. [18,19] calculated the values of pulses and pressures in the mechanically treated material using these assumptions. F. Delogu et al. [20,21] evaluated the effect of mill operating modes on the structure and phase composition of the mechanically treated mixture.

The equation for the mechanochemical reaction rate was derived using the Hertz theory [22,23]. A phenomenological kinetic model was used to describe mechanical treatment in a ball mill [24–26].

DEM and DEM-CDF calculations have been used to gain insight into mechanochemistry [27–29].

However, despite numerous experimental works and attempts to simulate mechanosynthesis, its mechanisms have not yet been reliably determined. A summary of the known results on mechanochemical synthesis processes can also be found, for example, in monographs and reviews [30–34]. Thus, the scientific literature offers different mechanochemical inorganic synthesis models, which include the parameters such as pressure, shear stress, deformation, and mass transfer.

Analysis of the above works reveals the major approaches to modeling mechanochemical reactions:

- the approach based on the thermal model. In this model, the major factor stimulating the reaction is heat release due to the friction of grinding bodies and particles of the treated substance [31,35–39];
- the approach using the theory of short-lived active centers appearing on the formed surface during milling [40,41];
- the approach including the formation of different defects due to plastic deformation during mechanical treatment [42–45].

The different theoretical approaches to the description of

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mechanochemical reactions complement each other. Combining the above approaches within the framework of the macrokinetic theory developed by us, we can restore the complete picture of mechanochemical synthesis in multicomponent systems.

Several key factors affect chemical reactions and the yield of the final product with desired characteristics [46,47]. One of the factors affecting the rate of chemical transformations is the kinetic factor, which is determined by the excess energy absorbed by various defects formed in the mechanically treated substance [48,49].

Another important factor is the structural factor, which is determined by the grinding of the reactants and the formation of an interfacial surface [50].

Intensive mechanical treatment leads to the heating of a reacting mixture and accelerates the chemical reaction. At the same time, dissipative heat release occurs due to the heating of the grinding bodies and mill walls. The factor that takes into account the heating of the treated substance and dissipative heat release is the temperature factor that accelerates the chemical reaction [51].

The chemical reaction equation should take into account all of the above factors. At the same time, a complete theoretical description of mechanochemical synthesis should include equations that describe the dynamics of excess energy in the treated substance, reaction surface area, temperature of the reacting mixture, grinding bodies, and the walls of the high-energy mill [46,52–56]. Here, the macroscopic mathematical models can be used to investigate the multiparameter problems of mechanochemical conversions in heterogeneous systems.

It should be noted that the majority of experimental data has been received only for the macroscopic analysis of processes. In particular, temperature measurements, chemical and X-ray diffraction analysis are performed on large volumes of material and the obtained values are macroscopic. Such averaged values are the consequence of macroscopic mathematical modeling.

The aim of this work is to study theoretically and experimentally the effect of mechanochemical treatment on the phase composition, dispersity and yield of the final product. The final product in the experimental studies was spinel cobalt ferrite nanopowder.

2. Mathematical model

The schematic of the mechanical treatment (MT) of the reacting powder mixture in the mill is shown in Fig. 1. The starting mixture consists of reactants R_1, R_2, and inert substance N (a). The activation and dispersion of the substances together with their agglomeration and the formation of the reaction product (b - d) take place during the MT process.

The reaction product consists of two substances: F (basic) and N

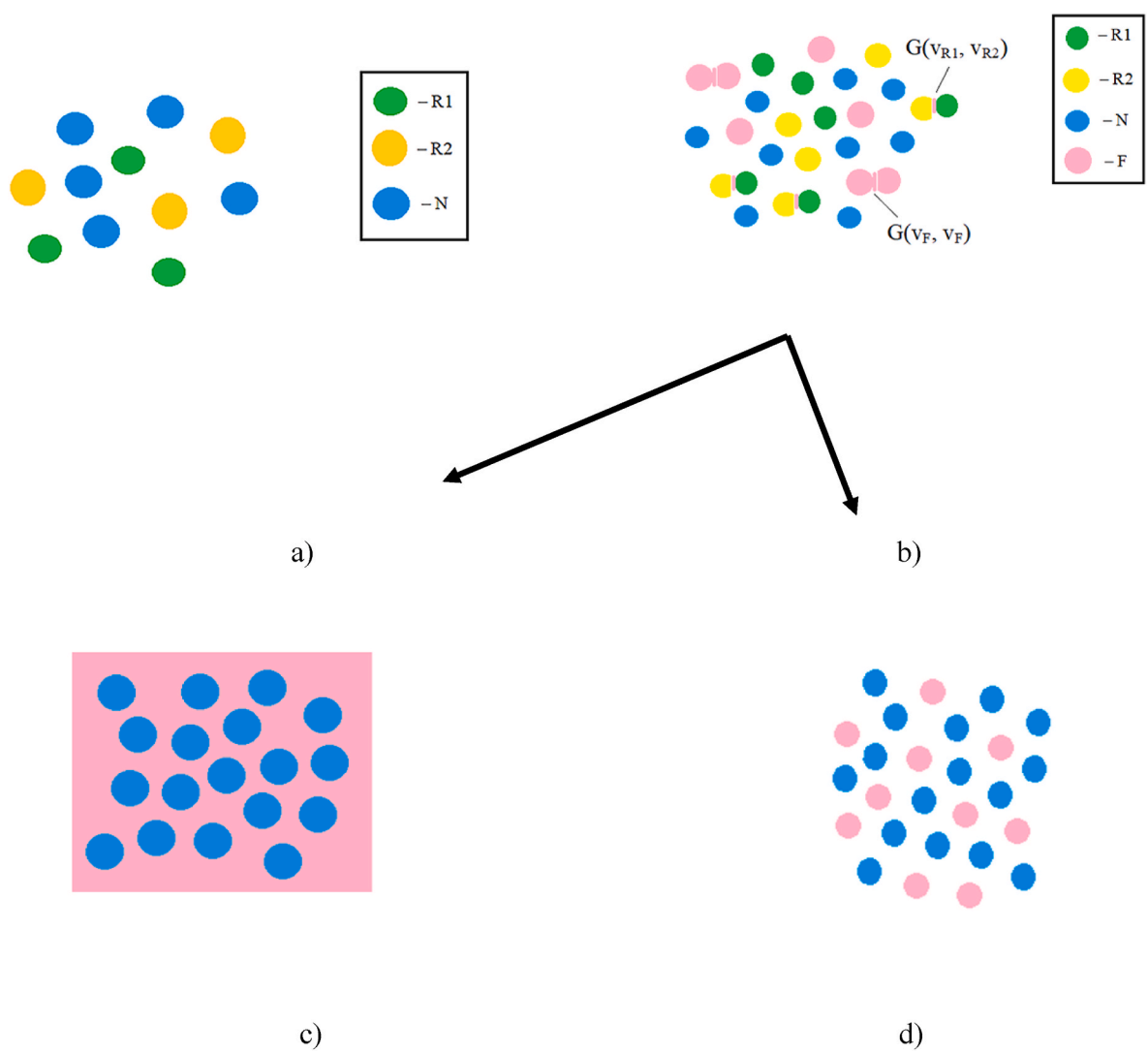

Fig. 1. Structure of the reaction powder mixture during mechanical treatment: (a) starting mixture; (b) grinding and activation of the mixture; (c) reaction product as a $F + N$ composite; (d) reaction product as ultradispersed $F + N$ particles.

(additive). Chemical interaction in the system proceeds according to the following formula:

$$R_1 + R_2 + N \rightarrow (F + N) + N.$$

Note that, depending on the mechanical treatment conditions that determine the mode of mechanochemical synthesis, the reaction product will be either a composite consisting of large agglomerates $F + N$ (c) or an ultrafine powder $F + N$ (d).

2.1. Governing equations of the mathematical model

Following the above schematic, the macroscopic description of mechanochemical synthesis in a three-component mixture is represented by the following system of equations taking into account structural, temperature, and kinetic factors [54–57]:

-grinding

$$h = \frac{h_0}{h_0 + (1 - h_0)exp(-k_{mill}t)}, \tag{1}$$

-dynamics of excess energy in reactants, additive, and product

$$\frac{d\varphi_i}{dt} = I_i - \varphi_i z_i \, exp\left(-\frac{U_i - \varphi_i}{RT}\right) \; (i = R_1, R_2), \tag{2}$$

$$\frac{d\varphi_N}{dt} = I_N - \varphi_N z_N \, exp\left(-\frac{U_N - \varphi_N}{RT}\right) - \frac{\varphi_N}{\overline{v}_N}\frac{d\overline{v}_N}{dt}, \tag{3}$$

$$\frac{d\varphi_F}{dt} = I_F - \varphi_F z_F \, exp\left(-\frac{U_F - \varphi_F}{RT}\right) - \frac{\varphi_F}{v_F}\frac{dv_F}{dt}, \tag{4}$$

- chemical transformation describing the rate of formation of the final product F during a one-step irreversible brutto-reaction

$$\frac{d\alpha}{dt} = K(T)f(\alpha)G(S), \tag{5}$$

-heat balance

$$\sum_{i=c,m,w} v_j c_j \rho_j \frac{dT}{dt} = Q\frac{dm_F}{dt} - \chi S_\Sigma (T - T_0). \tag{6}$$

The designations taken in (1) - (6) are as follows: t is the time; T is the temperature; R is the universal gas constant; $h = H/H_m$ is the dimensionless function that determines the change in particle size due to the grinding of components; $h_0 = H_0/H_m$ is the value of h at $t = 0$; H, H_m is the current and maximum particle size, respectively; k_{mill} is the grinding constant; φ_i, z_i, U_i, I_i are accumulated excess energy, pre-exponential factor, activation energy of excess energy relaxation, and accumulation rate of excess energy for the i-th component; $\varphi = \sum_{i=A,B,N,F} v_i\varphi_i$ is excess energy contained in a unit volume of powder mixture; v_i is the relative volume fraction of the i-th component in the powder mixture of volume V; $\overline{v}_N$ is the volume fraction of product N produced in reaction; α is the degree of mechanochemical conversion defined as the ratio of the mass of the starting reactants to the mass of the produced phase F (excluding the mass of additive N); m_F is the mass fraction of product F in the powder mixture; $k(T) = k_0 \exp\left(-\frac{E}{RT}\right), k_0, E = E_0 - [c_0\varphi_{R1} + (1 - c_0)\varphi_{R2}]$ are the constant, pre-exponential factor, and activation energy of the chemical reaction, accounting for the effect of excess energy stored in the reactants; E_0 is the activation energy of the chemical reaction in the absence of structural defects; c_0 is the mass fraction of reactant R1 in the reaction product; $f(\alpha)$ is the kinetic law of reaction; v_j, c_j, ρ_j are the volume fraction, heat capacity and density of the powder mixture (j = c), grinding bodies (j = m), and mill walls (j = w) in the mill chamber of volume V_Σ; S_Σ is the surface area of the mill chamber; Q is the heat effect of reaction per unit mass of product F; T_0 is the ambient temperature; χ is the coefficient of heat exchange of the mill chamber with the environment. It was assumed that the mechanical treatment constants used in (1) - (4) are directly proportional to the energy intensity of the mill, i.e., $I_i = a_i W/V$, (a_i is coefficients, W is the energy intensity of the mill, $i = R_1, R_2, N, F$). Grinding equation (1) is derived from a generalized grinding law which accounts for the energy of the mill spent on particle dispersion and surface formation [57]. In the equations for the rate of excess energy (2) - (4), the first term is responsible for the rate of accumulation of excess energy in the components, the second term is responsible for the relaxation of excess energy, and the third term in (3) and (4) accounts for the change in excess energy in N and F due to the chemical reaction. Heat balance equation (6) is written for the case of no temperature distribution in the grinding bodies and mill walls, which occurs when the time of thermal relaxation in the grinding bodies and mill walls is less than the characteristic time of chemical and dissipative heat release. The first term in this equation determines the rate of heat release during the formation of the reaction product, and the second term takes into account the heat transfer between the mill and the environment. This equation neglects the heating rate of the system due to the movement of the powder mixture and grinding bodies, and the heat release due to the change in excess energy in the activated components.

According to Ref. [54], the interfacial surface area in equation (5) can be represented by the relation

$$G(S) \approx SG(v_{R1}, v_{R2}), \tag{7}$$

where $S = 1/h$ is a dimensionless function that determines the specific surface of the particles. The function $G(v_{R1}, v_{R2})$ in (7) determines the effect on the formation of the interfacial surface between the reactants R_1 and R_2, taking into account their volume fractions v_{R1} and v_{R2}. In the first approximation, the function will have the form

$$G(v_{R1}, v_{R2}) \approx K_{R1,R2} v_{R1} v_{R2}. \tag{8}$$

From the physical point of view, $G(v_{R1}, v_{R2})$ characterizes the probability of formation of an interfacial surface between two heterogeneous reactants in a three-component system.

By analogy with the function $G(v_{R1}, v_{R2})$, the probability of agglomeration between the particles of the synthesized products in powder mixture $A + B + N + F$ can be represented by the relation as follows [54]

$$G(v_F, v_F) \approx K_{F,F} v_F^2. \tag{9}$$

The function $G(v_F, v_F)$ characterizes the probability of surface formation between the synthesized products. In (8), (9), $K_{R1,R2}$ and $K_{F,F}$ are coefficients.

Assuming that the value $p = 1 - G(v_F, v_F)$ corresponds to the probability of formation of the product F, then its yield can be estimated from the relation

$$\Psi = p\, m_F. \tag{10}$$

The value Ψ is some effective parameter used to optimize mechanosynthesis by stopping the mechanical treatment under controlled conditions at the time when Ψ becomes maximum.

In accordance with the notations introduced above, after simple but cumbersome transformations, the relations for the mass fractions of the components in the mixture can be represented in the form:

$$m_{R1} = c_0(1 - \nu)(1 - \alpha), m_{R2} = (1 - c_0)(1 - \nu)(1 - \alpha), m_N = \nu + \alpha\mu(1 - \nu), \\ m_F = \alpha(1 - \mu)(1 - \nu). \tag{11}$$

For the relative volumes of the components, the formula can be written as

$$v_i = \frac{m_i/\rho_i}{\sum_i (m_i/\rho_i)}, \bar{v}_N = \frac{\alpha\mu(1-\nu)/\rho_N}{\sum_i (m_i/\rho_i)}, (i = R_1, R_2, N, F). \quad (12)$$

In (11), (12) the designations are as follows: ν is the mixture ratio (initial mass of the additive relative to the initial mass of the mixture); μ is the mass fraction of N in the reaction products $F + N$ without considering N in the starting mixture; ρ_i is the density of the i-th component.

The initial conditions are written in the form

$$t = 0 : T = T_0, \alpha = 0, \varphi = \varphi_i = 0, h = h_0, S = \frac{1}{h_0}. \quad (13)$$

Thus, all equations, auxiliary relations, and initial conditions that constitute a macroscopic mathematical model of the mechanosynthesis of nanopowders have been determined.

2.2. Solution of the problem and analysis of results

In order to generalize the results obtained, highlight the main ones, and make the analysis convenient, we introduce the dimensionless parameters traditionally used when describing non-isothermal synthesis and dispersion of powder particles in a high-energy mill [54,55,58] as follows: $Ar = RT_*/E$ is the sensitivity of the reaction to temperature changes (Arrhenius number); $Td = c\rho T_*/(EQ)$ is the contribution of heat release from the chemical reaction (Todes number); $Se = \rho c V_\Sigma / (t_* \chi S_\Sigma)$ is the ratio between the characteristic time of heat exchange of the mixture with the environment $(\rho c V_\Sigma/(\chi S_\Sigma))$ and the characteristic time of the chemical reaction $\left(t_* = \exp\left(\frac{E_0}{RT_*}\right)\Big/ k_0\right)$ at the temperature T_* (Semenov number); $k^*_{mill} = k_{mill} t_*$ is the parameter that define the ratio of characteristic times of grinding $\left(\frac{1}{k_{mill}}\right)$ and chemical reaction; $\Delta_i = I_i / (V_\Sigma R T_*)$ characterizes the ratio between the rate of excess energy accumulation in the component i and the rate of chemical reaction; $P = \sum_{i=c,m,w} v_j c_j \rho_j / (V_\Sigma c \rho)$ is the parameter that reflects the characteristics of the mill and the grinding bodies. Dimensionless complexes reduce the number of parameters describing the problem. In the calculations, the ambient temperature ($T_* = T_0$) was taken as the scale temperature.

We also assume that the equality of thermophysical and physicochemical characteristics of all condensed substances in the powder mixture is fulfilled. This assumption makes it possible, among other things, not to consider the movement of the substance (volume changes) due to chemical transformations, which significantly simplifies the mathematical manipulations. The parameters, which did not vary in calculations, were as follows: $Ar = 0.08$, $Td = 0.1$, $P = 1$, $k^*_{mill} = 0.001$, $h_0 = 100$, $h_n = 1.2$, $K_{R1,R2} = 1$, $K_{F,F} = 5$, $\mu = 0.3$. Additionally, we believe that during mechanical treatment, the rate of relaxation of structural defects is negligibly small compared to the rate of their accumulation. Among the variety of kinetic laws describing the dynamics of chemical reactions [59], let us take the simplest form and assume the first order of the kinetic law of chemical transformations: $f(\alpha) = 1-\alpha$.

Fig. 2 shows the dynamics of temperature (a), the degree of mechanochemical conversion (Fig. 2, b, lines 1 - 3), and the amount of excess energy accumulated in the components (Fig. 2, b, lines 1' - 3′) during the mechanical treatment of a powder mixture in a high-energy mill. The values of the *Se* number, which characterizes the rate of heat transfer from the mill to the environment, varied.

It can be seen that as the *S*e number decreases (with intensification of the heat sink), an explosive mechanochemical synthesis (Fig. 2, a, lines 1 and 2) turns into structural and chemical transformations under practically isothermal conditions (a, line 3), which can be controlled by stopping the mechanical treatment at any convenient time.

As the Se number decreases, the time of chemical transformations increases (Fig. 2, b, lines 1 - 3), and structural changes in the mixture, associated with the accumulation of excess energy, occur faster due to the slower normalization of the structure of the substance during product formation (Fig. 2, b, lines 1' - 3′). Such "mild" conditions of synthesis allow avoiding high temperatures, which contributes to producing nanopowders from the final products [6].

The value of the Semenov number can be controlled, for example, by varying the dimensional and thermophysical characteristics of the mill chamber. However, in the experiment, for obvious reasons, this will be difficult to implement. Another way of controlling the Semenov number can provide the thermal explosion limit. For example, heat transfer in the reaction mixture can be achieved by intensive cooling of the operating mill. Intensification of the heat transfer can be achieved by reducing the ambient temperature.

To prevent an undesirable temperature rise in the activated mixture, the mode of the mechanical treatment process has to be switched from continuous to discrete. In the discrete mode, the mill operates with stops [60]. In this case, it is assumed that during the stop ("rest") of the mill, the heat accumulated in the activated substance during the mechanical

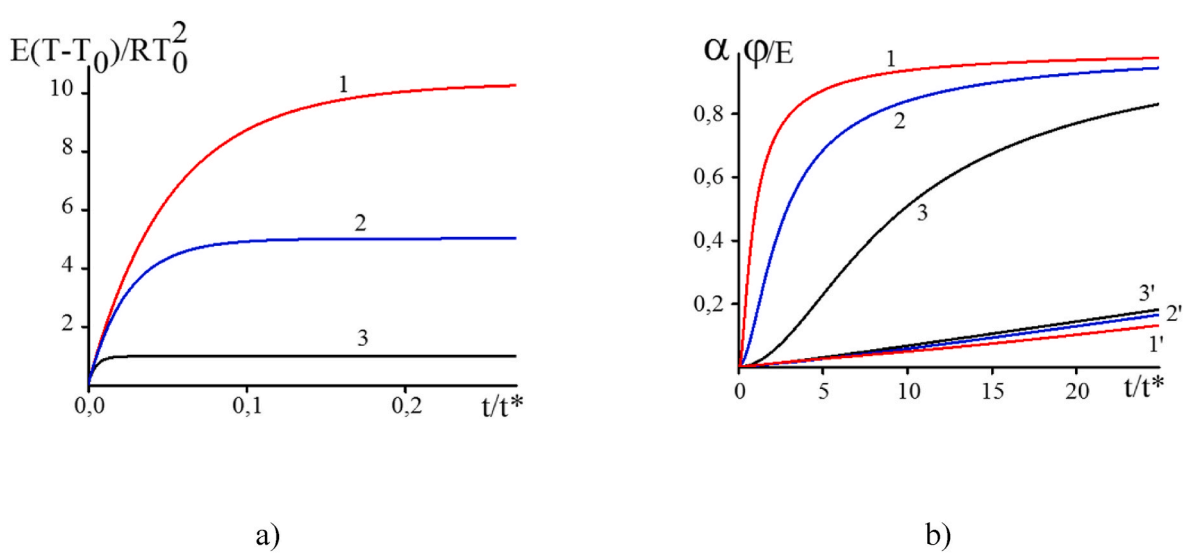

Fig. 2. Kinetic dependences of the temperature (a), the completeness of the reaction (b, lines 1 - 3), the amount of excess energy (b, lines 1' - 3′) during the mechanical treatment of the reaction mixture in a high-energy mill at $v_\Sigma = 0.01$, $\nu = 0.4$ and different Semenov number, Se: 1 - 10000, 2 - 5000, 3 - 1000.

treatment is removed. The operation and "rest" of the mill under conditions of the thermal explosion limit, which will depend to a greater extent on the number *Se*, can be estimated by the proposed mathematical model.

The dependencies of the fine product yield Ψ (lines 1 - 3) and the particle size (line 4) on the mechanical treatment time at different *Se* numbers and concentrations of the additive ν in the starting mixture are shown in Fig. 3. The horizontal and vertical lines indicate the shaded area, which defines the modes of producing nanoproduct *F* (particle size $h < h_n$). Here h_n is the value characterizing the nanoscale.

Analyzing Fig. 3, we can state that reaching the thermal explosion limit (small *Se*) by explosive mechanochemical synthesis (large *Se*) increases the yield of nanoproducts and shifts the maximum point of function Ψ(t) toward the shaded area (Fig. 3, a).

Note that the heat transfer in the system, characterized by the Se parameter, is only a necessary condition for producing nanopowder. This condition provides the possibility of achieving the thermal explosion limit (controlled soft mechanosynthesis mode). The addition of an inert component to the mixture, which reduces its reactivity, additionally contributes to this.

The addition of an inert to the starting mixture to prevent particle agglomeration also increases the yield of nanoproducts produced under controlled mechanical treatment conditions (Fig. 3, b). A small concentration of an inert component in the starting mixture does not provide a large yield of synthesized nanoparticles (Fig. 3b, line 1). A large concentration of the inert component increases the number of synthesized nanoparticles (Fig. 3b, lines 2 and 3).

Fig. 4 shows the mixture "ratio (ν) – mechanical treatment time (t_{mech})" (a) and "volume fraction of the starting powder mixture ($v_{\sum}$) - mechanical treatment time (t_{mech})" (b) diagrams, which reflect different scenarios of mechanochemical synthesis in the high-energy mill. Line 1 is the geometric locus of the parameters, which provide the maximum yield $\Psi = \Psi_m$ of the dispersed product *F*. Curve 2 is the geometric locus of the same parameters, at which non-sintered particles (or particles in the agglomerate) becomes nanoparticles h_n. Vertical line 3 passes through the intersection point of curves 1 and 2.

In region I, the particles of the mixture subjected to mechanical treatment are not nanoparticles ($h > h_n$). The modes of mechanochemical synthesis to produce nanoparticles ($h \le h_n$), correspond to region II cut off by lines 1, 2 and 3. The maximum yield of dispersed nanoproduct ($\Psi = \Psi_m$) will be observed in the areas adjacent to the straight line 1. As can be seen, the introduction of the additive reduces the probability of agglomeration and preserves the nanoparticles of the product (a). As the volume fraction of the powder mixture in the mill chamber increases, the duration of mechanochemical synthesis of nanoproduct particles decreases (b). In this case, the grinding efficiency decreases and the required particle size is achieved with longer times of mechanochemical treatment.

Let us obtain some analytical estimates. For this purpose, we introduce t_n and t_m, which are the times of controlled mechanical treatment of the powder mixture (reaching the thermal explosion limit).

Then, using the above equations of the mathematical model and omitting intermediate transformations, we finally write the relations for the parameters t_n and t_m

$$t_n = \frac{v_\Sigma}{k_{mill}} ln\left[\frac{h_n}{h_0}\left(\frac{1-h_0}{1-h_n}\right)\right], \tag{14}$$

$$t_m = \frac{RT_0}{I} v_\Sigma \, ln\left\{1 + \frac{1}{2}\left[\frac{1}{(1-\alpha_m)^2} - 1\right] \frac{It_*}{(1-\nu)^2 c_0 (1-c_0) K_{R1.R2} v_\Sigma RT_0}\right\}. \tag{15}$$

In (15), the value

$$\alpha_m = \alpha(t_m) = \begin{cases} \dfrac{1}{\sqrt{3K_{F,F}}(1-\mu)(1-\nu)}, \nu < \nu^* \\ 1, \nu \ge \nu^* \end{cases}.$$

characterizes mechanochemical transformation when reaching the maximum yield of the nanoproduct at time t_m. Here $\nu^* = 1 - 1/[\sqrt{3K_{F,F}}(1-\mu)]$.

As follows from (14) and (15), by varying the grinding (k_{mill}) and treatment (I) parameters, which determine the mechanochemical synthesis (for example, by changing the energy intensity of the mill), as well as the volume fraction of the mixture ($v_{\sum}$) and the concentration of the inert substance in the starting mixture (ν), we achieve a condition for nanoproduct formation, which is expressed as: $t_n \le t_{mech}$. In this case, the maximum yield of nanoproduct will be observed if $t_n \le t_{mech} = t_m$.

In turn, after simple transformations, the relationship for the maximum yield of powder nanoproduct *F* can be represented as follows

$$\Psi_m = \begin{cases} \dfrac{2}{3\sqrt{3K_{F,F}}}, \nu < \nu^* \\ \left[1 - K_{F,F}(1-\mu)^2(1-\nu)^2\right](1-\mu)(1-\nu), \nu \ge \nu^* \end{cases}. \tag{16}$$

Relation (16) optimizes the production of a powder nanoproduct in terms of the parameter ν. For this purpose, examining the function $\Psi_m(\nu)$ for increase and decrease, we find that the value of Ψ_m at $\nu \le \nu^*$ is larger than at $\nu > \nu^*$. Therefore, we can add another condition $\nu \le \nu^*$ to the

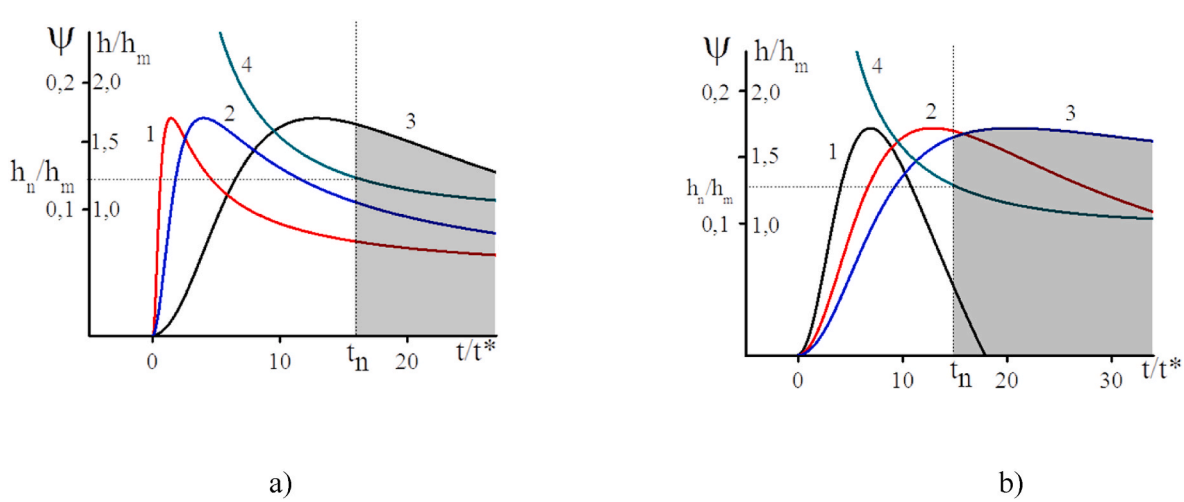

Fig. 3. Kinetic dependences of the product yield (lines 1 - 3) and the relative particle size (line 4) on the Se (a) and ν (b) parameters at $v_\Sigma = 0.01$; (a): $\nu = 0.4$, (1) Se = 10000, (2) Se = 100, (3) Se = 40; (b): Se = 1000, (1) $\nu = 0.2$, (2) $\nu = 0.4$, (3) $\nu = 0.6$. The rest parameters correspond to the caption to Fig. 2.

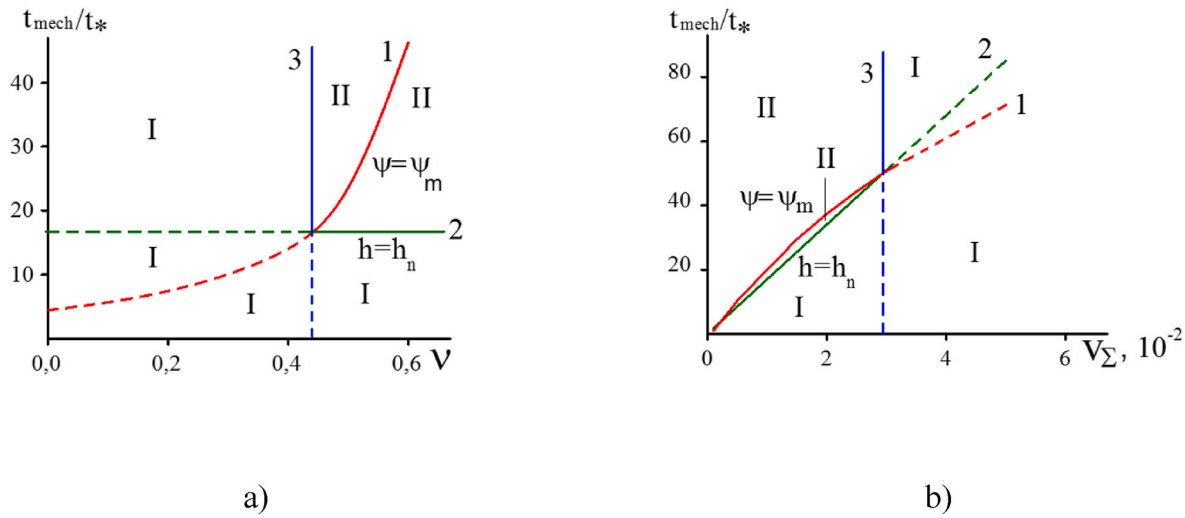

Fig. 4. "Inert substance concentration – mechanical treatment time" diagram (a) at $v_\Sigma = 0.01$, Se = 1000 and "volume fraction of starting powder mixture - mechanical treatment time" diagram (b) at $\nu = 0.5$, Se = 1000. The rest of the parameters correspond to the caption to Fig. 2.

optimization of nanopowder mechanosynthesis.

3. Experiment

3.1. Experimental procedure

The theoretical results were compared with the experimental data on the mechanochemical synthesis of spinel cobalt ferrite nanopowders. The reaction was as follows:

$$2FeCl_3\text{-}6H_2O + CoCl_2\text{-}6H_2O + Ca(OH)_2 + 3Na_2CO_3 = Co_xFe_{3-x}O_4 + 6NaCl + 3CO_2\uparrow + 13H_2O$$

Starting materials for the synthesis were chemically pure reagents, and iron and cobalt chlorides in the reaction were crystalline hydrates. To prevent heating of the reactant mixture (r.m.) and aggregation of the final product during mechanochemical synthesis, an inert component, sodium chloride, was introduced into the reactant mixture with different ratios: $m_{r.m.}$: $m_{NaCl} = 1{:}1 \div 1{:}4$.

Sodium chloride in mechanochemical reactions, in our and other researchers' opinion who first proposed the use of the salt matrix in mechanochemistry [61] is very useful. The major advantage of sodium chloride over other inert substances is that it dissolves in water and is easily separated from the reaction products. The complete removal of chloride ions is controlled by a qualitative reaction between chloride ions and silver nitrate.

A mixture of reactants and an inert component was placed into steel jars (drums) in an MPV planetary mill. Steel balls with a diameter of 5 mm were added to the mixture at the ball-to-powder ratio m_b: $m_p = 10{:}1$. Discrete mechanochemical synthesis was performed for different treatment times $\tau = 5 : 60$ min, with stops every 5 min for cooling of the mill jars. The values m_b and m_p used in the experiments are related to the parameter ν in the mathematical model by the following relation $\nu = m_b/(m_b + m_p)$.

The synthesized product was washed with distilled water until the salts were completely removed (20 °C). Then, if necessary, it was treated with ultrasound and centrifuged. Impurity chlorine ions were controlled by a qualitative reaction between chloride ions and silver nitrate. The final product was thermally treated at 100 °C in air. The phase composition, morphology, and dispersion of the synthesized powders were characterized by X-ray diffraction (Schimadzu XRD 6000, CuK radiation) or electron microscopy (EM-125 multimeter). The XRD data were processed using the analysis program (POWDER CELL 2.5).

3.2. Results and discussion

The final product of mechanochemical synthesis is a light-brown spherical nanopowder. As can be seen from Table 1, which presents the average particle diameter, with an increase in the treatment time from 5 to 60 min, the average particle size changes slightly. These results are in good agreement with the data determined earlier by X-ray diffraction and electron microscopy in Refs. [62,63].

Calculations showed that the grinding of the mixture occurs during the period from 0- to 1.5-min mechanical treatment (Fig. 5). When the grinding limit is reached, the particle size practically does not change. In this case, the computed calculation is in good agreement with the experimental data (Table 1) for the grinding parameters included in (1) $k_{mill} = 1$ min^{-1} and $H_m = 8.33$ mμ.

Fig. 6 shows X-ray diffraction patterns of spinel cobalt ferrite powders. The X-ray diffraction patterns show high-intensity peaks of sodium chloride caused by its excessive concentration in the reaction system (a). The complete removal of the salt matrix ions is confirmed by the X-ray diffraction patterns (b) and (c), which show only the diffraction lines of the mechanochemical reaction products.

The synthesized nanopowders are heterophase and consist of spinel cobalt ferrite (the main phase), β-modified iron monohydrate β-FeO(OH), small amounts of α-Fe_2O_3 iron oxide, and an amorphous phase. With increasing the duration of mechanochemical synthesis, the concentration of spinel cobalt ferrite in the final product increases, reaching 90% at the treatment time of 25-30 min, after which it changes weakly (Table 2).

Let us make a correlation between the experimental data on the

Table 1
Particle size of spinel cobalt ferrite as a function of the mechanical treatment time. Air, $m_{r.m.}$: $m_{NaCl} = 1{:}4$, m_p: $m_b = 1{:}20$.

Time of mechanochemical treatment, min	Average particle size, nm
0	3.73×10^6
5	7
10	8
15	8
20	9
25	10
30	10
35	11
45	11
50	10
60	10

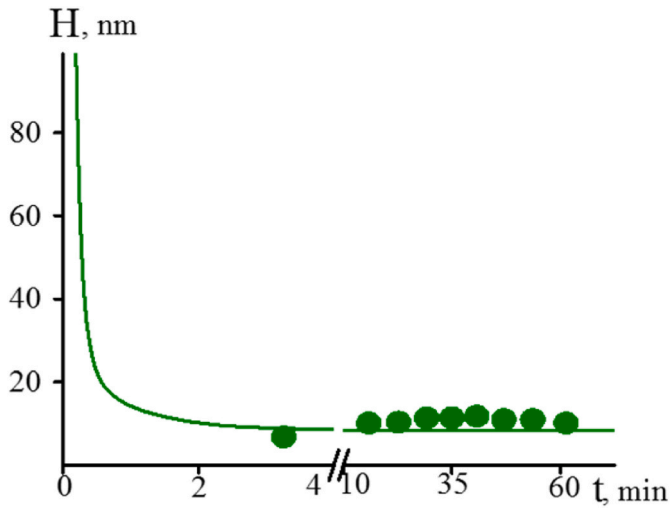

Fig. 5. Theoretical (solid line) and experimental (dots) dependences of the particle size of spinel cobalt ferrite on the mechanical treatment time at $\nu = 0.8$ (or air, $m_{r.m.}$: m_{NaCl} = 1:4).

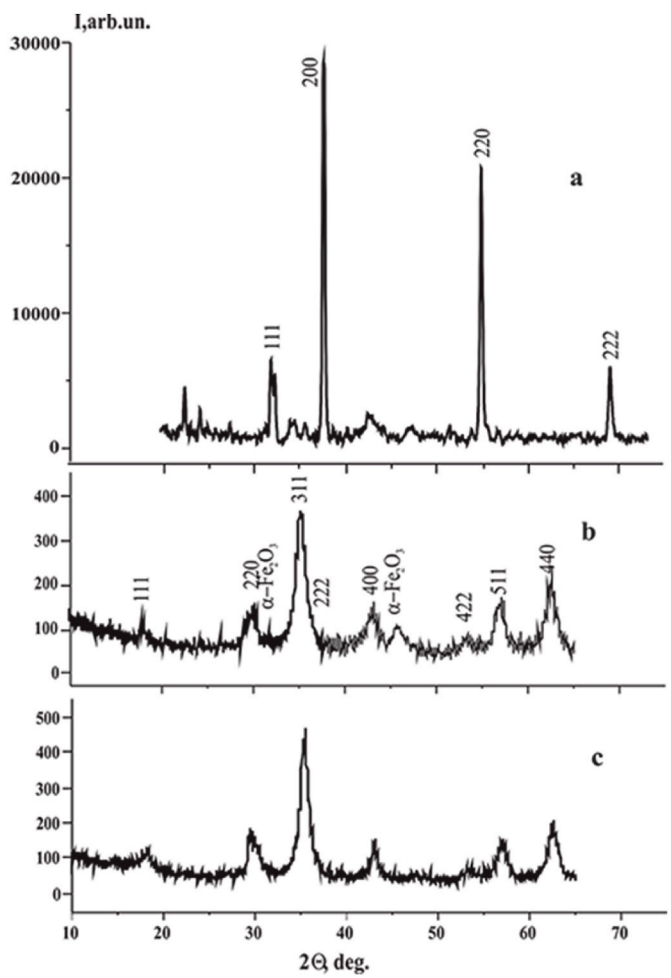

Fig. 6. X-ray diffraction patterns of spinel cobalt ferrite nanopowders after: mechanical treatment (a), washing from salt matrix (b), heat treatment at 100 °C (c).

mechanochemical transformation of the starting substances into spinel cobalt ferrite and the theoretical estimates obtained using the mathematical model proposed above. For this purpose, integrating the equation of chemical transformation (5) considering (7) and (11) in isothermal approximation (when mechanosynthesis of the product is performed under controlled limit of thermal explosion), we finally find the following analytical formula for the mechanochemical transformation:

$$\alpha = 1 - \sqrt{\frac{1}{1 + 2(1-\nu)^2 M[t + P(t)]}}, \tag{17}$$

$$M = K(T)K_{R1,R2}c_0(1-c_0), P(t) = \frac{[\exp(-k_{mill}t) - 1]}{k_{mill}}.$$

Table 2
Concentration of spinel cobalt ferrite in the final product as a function of the mechanical treatment time. Air, $m_{r.m.}$: m_{NaCl} = 1:4, m_p: m_b = 1:20.

Time of mechanochemical treatment, min	Spinel cobalt ferrite concentration, wt%
0	0
5	37
10	66
15	76
20	88
25	90
30	91
35	87
45	86
50	86
60	85

Fig. 7 shows the theoretical dependence of mechanochemical conversion of the starting mixture into spinel cobalt ferrite on the mechanical treatment time. It can be seen that in the period up to 10-min mechanical treatment, there is a rapid increase in the degree of mechanochemical conversion up to the value α ~0.8. However, then, the degree of mechanochemical conversion decreases, and the value of α changes weakly. For M = 0.94 min^{-1} and ν = 0.8, the theoretical calculation by formula (17), derived using the kinetic law $f(\alpha) = 1-\alpha$, is in good agreement with the experimental results presented in Table 2.

The concentration of the inert additive (NaCl) in the system significantly affects the conversion degree of chemical reactants during the process of mechanosynthesis. When the ratio $m_{r.m.}$:m_{NaCl} = 1:1 and the time of mechanochemical synthesis is 30 min, the degree of conversion is low and the yield of the final product is less than 60 wt% (Table 3).

By increasing the NaCl concentration in the reaction mixture to the ratio $m_{r.m.}$:m_{NaCl} = 1:4, the yield of spinel cobalt ferrite increases up to 90 wt%. With a further increase in the concentration of the inert

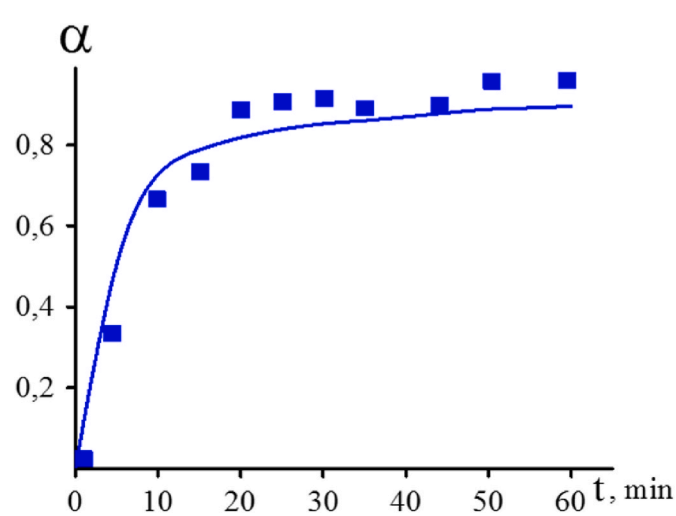

Fig. 7. Theoretical (solid line) and experimental (square dots) dependences of the mechanochemical conversion of the starting mixture into spinel cobalt ferrite on the mechanical treatment time at $\nu = 0.8$ (or air, $m_{r.m.}$: m_{NaCl} = 1:4).

Table 3
Phase composition of spinel cobalt ferrite as a function of the concentration of the inert additive (NaCl) in the reaction mixture. Mechanical treatment time = 30 min, m_p:m_b = 1:20.

Reactant-to-NaCl ratio	Phase composition, wt%
1:1	spinel – 58 Fe_2O_3 – 12 β-FeOOH – 30
1:2	spinel – 81 Fe_2O_3 – 4 β-FeOOH – 15
1:3	spinel – 84 Fe_2O_3 – 7 β-FeOOH – 9
1:4	spinel – 90 Fe_2O_3 – 3 β-FeOOH – 7
1:5	spinel – 82.08 Fe_2O_3 – 11.68 β-FeOOH – 6.24

component in the reaction mixture to 1:5 the yield of the final product decreases.

In this study, we could not find any reason for the non-monotonic dependence of spinel cobalt ferrite nanoparticles on the concentration of inert material ν. This dependence can be explained in terms of nanoproduct yield rather than the mechanochemical conversion of the starting mixture into a reaction product. Then, the dependence presented in Table 3 can be described by the product yield function Ψ, determined by formula (10). It can be seen that the theoretical calculation of the yield of spinel cobalt ferrite nanopowder by formula (10) at $K_{F,F}$ = 11.5 (Fig. 8) is in good agreement with the experimental data presented in Table 3. In Fig. 8, the value Ψ_m denotes the maximum possible yield of nanoproduct under these conditions of mechanical treatment.

The theoretical dependencies of mechanochemical conversion (curve 1) on the concentration of the inert additive in the starting mixture for the maximum yield of spinel cobalt ferrite nanopowder (curve 2) are shown in Fig. 9.

It can be concluded that increasing the concentration of the inert additive significantly increases the degree of conversion α_m with a practically unchanged maximum yield of the final product Ψ_m. In this case, when the concentration of the additive is ν>0.75, the maximum

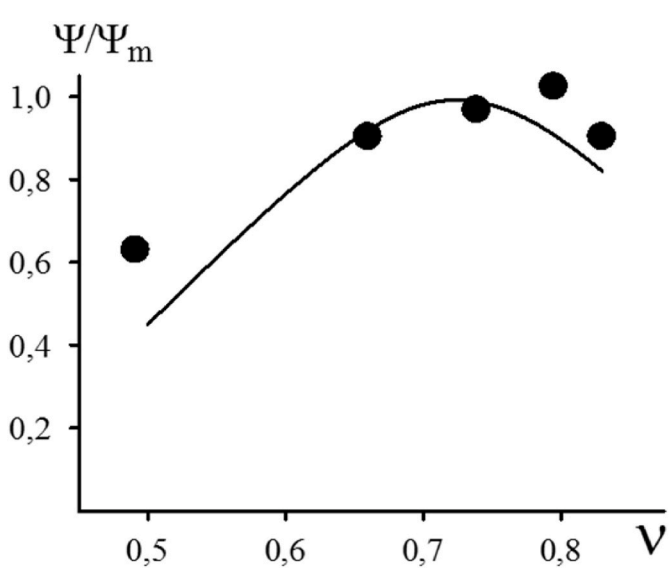

Fig. 8. Theoretical (solid line) and experimental (dots) dependences of the yield of the final spinel cobalt ferrite nanopowder on the volume fraction of the inert additive in the starting mixture, mechanical treatment time = 30 min.

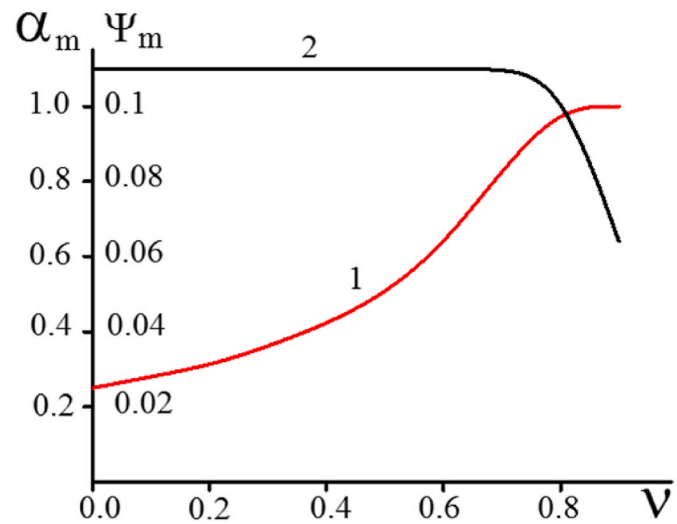

Fig. 9. Theoretical dependences of the values α_m (line 1) and Ψ_m (line 2) on the concentration of an inert component in the starting mixture. α_m is the degree of mechanochemical conversion of the starting mixture into spinel cobalt ferrite nanopowder with the maximum yield (Ψ_m).

yield of the final product begins to decrease. Thus, at small ν, a large number of semifinished nanoparticles are formed, and the degree of chemical conversion will be low. A large concentration of the inert additive in the mixture can be used to produce homogeneous nanoparticles of spinel cobalt ferrite, but in a small amount. Here, the choice of the optimal inert additive will be determined by the final goal of the mechanical treatment: either producing the large amount of nanopowder ($\alpha_m \ll 1$), or limited portions of nanopowder ($\alpha_m \sim 1$).

Using the proposed mathematical model and the determined kinetic constants, a theoretical prognostic diagram was plotted in the "mechanical treatment time (t_{mech}) - concentration of inert additive NaCl (ν)" coordinates. The diagram is divided into three main areas and defines the mechanochemical synthesis of spinel cobalt ferrite under controlled conditions (Fig. 10) allowing to stop the mill at any time.

In region I, the powder mixture is incompletely milled. Here, the powder mixture particles are not nanoparticles, and the degree of mechanochemical conversion is low. In region II, the powder mixture is well milled and the degree of mechanochemical conversion is high, but the yield of spinel cobalt ferrite nanopowder is small ($\Psi << \Psi_m$).

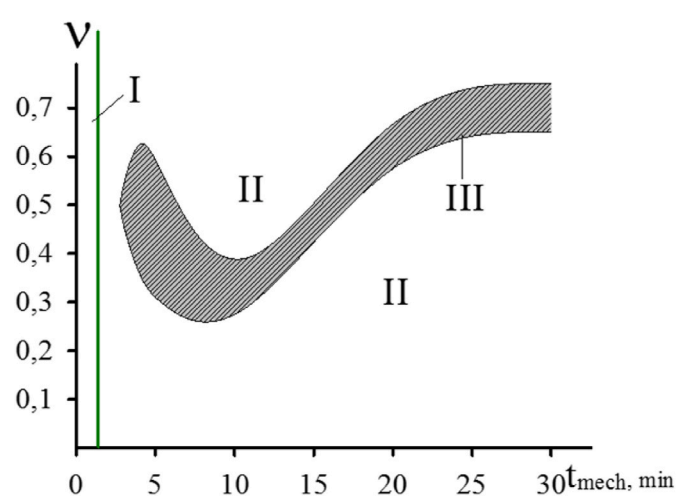

Fig. 10. Calculated "mechanical treatment time - concentration of inert substance (NaCl)" diagram, which determines different modes of mechanosynthesis of spinel cobalt ferrite nanopowder: incomplete grinding of the mixture (I), small (II) and maximum (III) yield of final nanopowder.

The optimal mechanochemical synthesis of the spinel cobalt ferrite nanopowder correspond to gray region III. Here, the yield of the final product is close to the maximum (Ψ∼Ψm). Analyzing the diagram, we can conclude that the effective values of the inert additive ν, leading to the maximum yield of the final product, correspond to 5-min mechanical treatment. As the mechanical treatment time of the mixture increases, the range of effective values of the inert additive significantly narrows, since the probability of agglomeration of synthesized spinel cobalt ferrite particles increases. The region of optimal modes is a V-shaped, with a minimum at $t = 9$ min mechanical treatment. After 25 min, further mechanical treatment is unproductive, since the range of effective values of the inert additive does not depend on the mechanical treatment time.

Thus, this paper demonstrates the possibility of producing nanopowders by mechanochemical synthesis. The macrokinetic model provides predictive estimates for controlling mechanochemical synthesis in three-component powder mixtures, one component of which is an inert additive. The algorithmized scheme of the study, which demonstrates the effect of different factors of mechanochemical treatment on mechanosynthesis of the final product, is presented in Fig. 11.

When comparing the experimental data with theoretical results, the following can be noted. In the present study, experiments were performed using a high-energy planetary mill. The discrete mode of the mill automatically provides the chemical process under the limit of thermal explosion at small Semenov numbers (*Se*), when chemical and structural transformations develop under "soft", practically isothermal conditions, rather than in an explosive mode. The addition of inert sodium chloride proposed in Ref. [14] to the reaction mixture to prevent agglomeration and reduce the influence of the temperature factor optimizes the mechanochemical synthesis of nanoparticles. At the same time, the yield of nanoparticles weakly increases with the increase in the concentration of the inert component in the reaction mixture.

It should also be noted that the experimental dependence of the nanoproduct yield on the time of mechanochemical treatment corresponds to the calculations obtained by the model. This testifies to the adequacy of the model representations to the experimental results.

4. Conclusions

A macroscopic mathematical model of the mechanochemical synthesis of nanopowders in a high-energy mill was developed. The different modes of mechanochemical treatment of a powder mixture were shown, including optimal modes with the maximum yield of nanoproducts. The synthesis of nanopowders can be optimized by:

- performing mechanosynthesis under controlled conditions (under the thermal explosion limit);
- adding an inert component to the starting reaction mixture;
- reducing the volume fraction of the starting mixture.

The theoretical results obtained are in good agreement with the experimental results. An analytical formula linking the degree of mechanochemical conversion with the time of mechanical treatment and the concentration of the inert component in the starting mixture was derived. The kinetic constants characterizing the mechanochemical synthesis of spinel cobalt ferrite nanopowder were determined using the inverse problem: the grinding rate constant (k_{mill}); the powder particle size corresponding to the grinding limit (*Hm*); the mechanochemical reaction rate constant (*M*); the factor determining the probability of agglomeration between the product particles ($K_{F,F}$).

Varying the conditions of mechanochemical treatment of the starting reaction mixture makes it possible to produce nanoparticles dispersed in a soluble salt matrix. Selective removal of the matrix phase by washing the synthesized powder with solvent provides a final product with a small particle size, narrow size distribution, and homogeneous crystal structure and morphology.

Credit author statement

Prof. O.V. Lapshin: Conceptualization, methodology, formal analysis, investigation, visualization, manuscript-original draft & editing.
Dr. A.A. Nevmyvaka: Experimentation, analysis, investigation.
Prof. V.I. Itin: Experimentation, analysis, investigation.
Dr. O.V. Ivanova: Writing—review and editing, data curation.

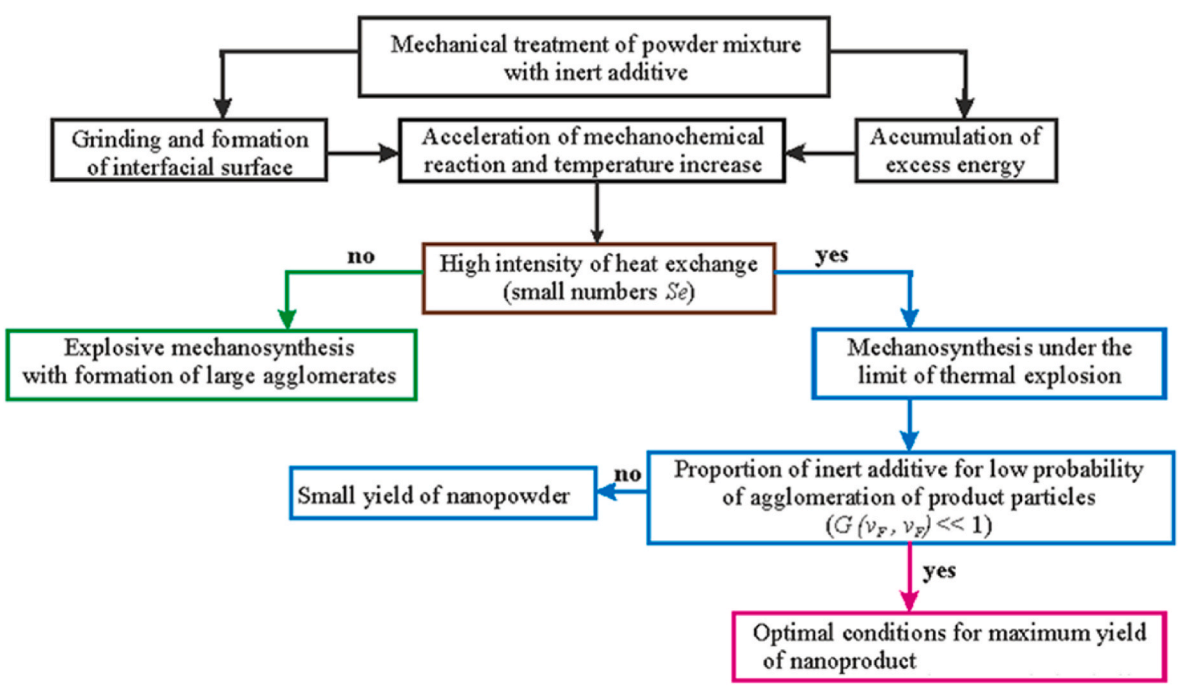

Fig. 11. Scheme of the mechanochemical synthesis of nanopowders in a high-energy mill.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] L. Takacs, The historical development of mechanochemistry, Chem. Soc. Rev. 42 (2013) 7649–7659, https://doi.org/10.1039/C2CS35442J.
[2] A.S. Rogachev, Mechanical activation of heterogeneous exothermic reactions in powder mixtures//Russ, Chem. Rev. 88 (2019) 875–900, https://doi.org/10.1070/RCR4884.
[3] Adam A.L. Michalchuk, Elena V. Boldyreva, Ana M. Belenguer, franziska emmerling and vladimir V. Boldyrev. Tribochemistry, mechanical alloying, mechanochemistry: what is in a name? Front. Chem. 9 (2021) https://doi.org/10.3389/fchem.2021.685789.
[4] F. Delogu, G. Gorrasi, A. Sorrentino, Fabrication of polymer nanocomposites via ball milling: present status and future perspectives, Prog. Mater. Sci. 86 (2017) 75–126, https://doi.org/10.1016/j.pmatsci.2017.01.003.
[5] M.A. Korchagin, T.F. Grigor'eva, B.B. Bokhonov, M.R. Sharafutdinov, A. P. Barinova, N.Z. Lyakhov, Solid-state combustion in mechanically activated SHS systems. I. Effect of activation time on process parameters and combustion product composition, Combust. Explos. Shock Waves 39 (2003) 43–50, https://doi.org/10.1023/A:1022145201911.
[6] M.A. Korchagin, T.F. Grigor'eva, B.B. Bokhonov, M.R. Sharafutdinov, A. P. Barinova, N.Z. Lyakhov, Solid-state combustion in mechanically activated SHS systems. II. Effect of mechanical activation conditions on process parameters and combustion product composition, Combust. Explos. Shock Waves 39 (2003) 51–58, https://doi.org/10.1023/A:1022197218749.
[7] Takuya tsuzuki «mechanochemical synthesis of nanoparticles, in: Book: Reference Module in Materials Science and Materials Engineering, 2022, https://doi.org/10.1016/B978-0-12-822425-0.00058-0.
[8] Said M. Al Azar, Ahmad A. Mousa, Mechanical and Physical Methods for the Metal Oxide Powders production//Metal Oxide Powder Technologies, 2020, pp. 169–187, https://doi.org/10.1016/B978-0-12-817505-7.00008-7.
[9] Ghada A. Al Bazedi, Aiman Eid Al-Rawajfeh, Mona A. Abdel-Fatah, Mohammad R. Alrbaihat, Ehab AlShamaileh, Synthesis of Nanomaterials by Mechanochemistry//Handbook of Greener Synthesis of Nanomaterials and Compounds, 2021, pp. 405–418, https://doi.org/10.1016/B978-0-12-821938-6.00011-6.
[10] at al. Matej Baláž, Mechanochemistry as a versatile and scalable tool for nanomaterials synthesis: recent achievements in Košice, Slovakia, Current Opinion in Green and Sustainable Chemistry 24 (2020) 7–13, https://doi.org/10.1016/j.cogsc.2019.12.007.
[11] Davin Tan, Mechanochemistry for organic chemists, Update/Eur. J. Org Chem. 1 (2018) 18–33, https://doi.org/10.1002/ejoc.201700961.
[12] José G. Hernández, Carsten Bolm, Altering product selectivity by mechanochemistry, J. Org. Chem. 82 (8) (2017) 4007–4019, https://doi.org/10.1021/acs.joc.6b02887.
[13] A.S. Mukasyan, B.B. Khina, R.V. Reeves, S.F. Son, Mechanical activation and gasless explosion: nanostructural aspects, Chem. Eng. J. 174 (2011) 677–686, https://doi.org/10.1016/j.cej.2011.09.028.
[14] T. Tsuzuki, P. McCormick, Mechanochemical synthesis of nanoparticles, J. Mater. Sci. 39 (2004) 5143–5146, https://doi.org/10.1023/B:JMSC.0000039199.56155.f9.
[15] H. Sharifi, S.R. Faradonbeh, M. Tayebi, Production and characterization of cobalt/vanadium boride nanocomposite powder by mechanochemical method//Mater, Chem. Phys. 202 (2017) 251–257, https://doi.org/10.1016/j.matchemphys.2017.09.015.
[16] J. Ding, W.F. Miao, P.G. McCormick, R. Street, Mechanochemical synthesis of ultrafine Fe powder, Appl. Phys. Lett. 67 (1995) 3804, https://doi.org/10.1063/1.115389.
[17] F. Delogu, G. Mulas, L. Schiffini, G. Cocco, Mechanical work and conversion degree in mechanically induced processes//mater, Sci. Eng. A. 382 (1–2) (2004) 280–287, https://doi.org/10.1016/j.msea.2004.05.047.
[18] F.K. Urakaev, V.V. Boldyrev, Mechanizm and kinetics of mechanochemical processes in comminuting devices. 1, Theory//Powder Technol. 107 (1–2) (2000) 93–107, https://doi.org/10.1016/S0032-5910(99)00175-8.
[19] F.K. Urakaev, V.V. Boldyrev, Mechanizm and kinetics of mechanochemical processes in comminuting devices. 2. Applications of the theory, Experiment//Powder Technol. 107 (3) (2000) 197–206, https://doi.org/10.1016/S0032-5910(99)00200-4.
[20] F. Delogu, Activation of self-sustaining high-temperature reactions by mechanical processing of Ti–C powder mixtures, Scripta Mater. 69 (2013) 223–226, https://doi.org/10.1016/j.scriptamat.2013.03.033.
[21] F. Delogu, L. Takacs, Mechanochemistry of Ti–C powder mixtures, Acta Mater. 80 (2014) 435–444, https://doi.org/10.1016/j.actamat.2014.08.036.
[22] F. Kh Urakaev, V.V. Boldyrev, Kinetics of mechanochemical processes in grinding machines, Inorg. Mater. 35 (4) (1999) 405–412.
[23] F. Kh Urakaev, T.A. Ketegenov, O.A. Tyumentseva, V.V. Boldyrev, A simulation of the interaction between the material of milling bodies and a substance under processing for the example of quartz processing in a mill with steel fittings, Russ. J. Phys. Chem. A 78 (5) (2004) 710–715.
[24] F. Delogu, L. Takacs, Information on the mechanism of mechanochemical reaction from detailed studies of the reaction kinetics, J. Mater. Sci. 53 (2018) 13331–13342, https://doi.org/10.1007/s10853-018-2090-1.
[25] G. Traversari, A. Porcheddu, G. Pia, F. Delogu, A. Cincotti, Coupling of mechanical deformation and reaction in mechanochemical transformations, Phys. Chem. Chem. Phys. 23 (2021) 229–245, https://doi.org/10.1039/D0CP05647B.
[26] M. Carta, E. Colacino, F. Delogu, A. Porcheddu, Kinetics of mechanochemical tranformations//Phys. Chem. Chem. Phys. 22 (2020) 14489–14502, https://doi.org/10.1039/D0CP01658F.
[27] A. Sato, J. Kano, F. Saito, Analysis of abrasion mechanism of grinding media in a planetary mill with DEM simulation, Adv. Powder Technol. 21 (2010) 212–216, https://doi.org/10.1016/j.apt.2010.01.005.
[28] Kim Kyong-Chol, Tao Jiang, Yingzhe Xue, Nam-il Kimf, Jing Hak-Son, Kimh Jong-Chol, Application of discrete element simulation in mechanical activation of boron concentrate//Powder, Technol. 382 (2021) 441–453, https://doi.org/10.1016/j.powtec.2020.12.031.
[29] Tamir Brosh, Haim Kalman, Avi Levy, Isabelle Peyron, Francois Ricard, DEM–CFD simulation of particle comminution in jet-mill, Powder Technol. 257 (2014) 104–112, https://doi.org/10.1016/j.powtec.2014.02.043.
[30] V.V. Boldyrev, Mechanochemistry in siberia//Her, Russ. Acad. Sci. 88 (2) (2018) 142–150, https://doi.org/10.7868/S086958731803012X.
[31] G. Heinicke, Tribochemistry. C, . Hanser Verlag, Munich,, 1985, p. 495.
[32] P. Yu Butyagin, A.N. Streletskii, The kinetics and energy balance of mechanochemical transformations//phys, Solid State 47 (35) (2005) 856–862, https://doi.org/10.1134/1.1924845.
[33] E. Avvakumov, M. Senna, N. Kosova, Soft Mechanochemical Syntthesis: A Basics for New Chemical Technologies, Kluwer Acad. Publ., Boston; Dordrecht; London, 2001, p. 201.
[34] B.B. Khina, Combustion Synthesis of Advanced Materials, Nova Science Publishers, New York, 2010, p. 110.
[35] F.P. Bowden, P.H. Thomas, The surface temperature of sliding solids, Proc. R. Soc. London, A 223 (1954) 29–40, https://doi.org/10.1098/rspa.1954.0098.
[36] P. Thiessen, G. Mayer, K. Heinike, Grundlagen der Tribochemie. Berlin, 1967, p. 194.
[37] P.G. Fox, J. Soria-Ruiz, Fracture-Induced thermal decomposition in brittle crystalline solids, Proc. R. Soc. London, A A. 317 (1970) 79–90, https://doi.org/10.1098/rspa.1970.0103.
[38] F. Kh Urakaev, V.V. Boldyrev, V.R. Regel, et al., Mechanism of mechanochemical decomposition of solid inorganic compounds, Kinet. Catal. 18 (1977) 350–358.
[39] R. Weichert, K. Schonert, On the temperature rise at the tip of a fast running crack, J. Mech. Phys. Solid. 22 (2) (1974) 127–133, https://doi.org/10.1016/0022-5096(74)90018-0.1.
[40] E.M. Gutman, Mechanochemistry of Materials, Materials Engineering Dept., Ben-Gurion University, Beer-Sheva, Israel, 1998, p. 215.
[41] P.Y. Butyagin, Forced reactions in inorganic and organic chemistry, Colloid J. 61 (5) (1999) 537–544.
[42] E. Ivanov, V. Neverov, N. Jitnikov, et al., Nucleation of phases in Ni-Al deformed in Bridgman anvils//Mater, Letture 7 (1–2) (1988) 57–60, https://doi.org/10.1016/0167-577X(88)90083-3.
[43] E. Ivanov, G.V. Golubkova, T.F. Grigorieva, The CsCl structure of the Raney nickel catalyst obtained from mechanically alloyed nickel aluminides, React. Solid 8 (1–2) (1990) 73–76, https://doi.org/10.1016/0168-7336(90)80009-9.
[44] Investigations and applications of severe plastic deformation, in: T.C. Lowe, R. Z. Valiev (Eds.), NATO Science Series. Series 3 80, High Technology, 2000, p. 394.
[45] R.Z. Valiev, R.K. Islamgaliev, I.V. Alexandrov, Bulk nanostructured materials from severe plastic deformation, Prog. Mater. Sci. 45 (2) (2000) 103–189, https://doi.org/10.1016/S0079-6425(99)00007-9.
[46] O.V. Lapshin, E.V. Boldyreva, V.V. Boldyrev, Role of mixing and milling in mechanochemical synthesis (Review), Russ. J. Inorg. Chem. 66(3) (2021) 433–453, https://doi.org/10.1134/S0036023621030116.
[47] Adam A.L. Michalchuk, Franziska Emmerling, Time-resolved in situ monitoring of mechanochemical reactions (minireview), Angew. Chem. Int. Ed. 61 (21) (2022), e202117270, https://doi.org/10.1002/anie.202117270.
[48] Mohammad Alrbaihat, Firas Khalil Al-Zeidaneen, Qusay Abu-Afifeh, Reviews of the kinetics of mechanochemistry: theoretical and modeling aspects, Mater. Today Proc. 65 (8) (2022) 3651–3656, https://doi.org/10.1016/j.matpr.2022.06.195.
[49] at al. EvaGil-González, Unveiling mechanochemistry: kinematic-kinetic approach for the prediction of mechanically induced reactions, J. Alloys Compd. 866 (2021), 158925, https://doi.org/10.1016/j.jallcom.2021.158925.
[50] Pavel Yu Butyagin, Structural disorder and mechanochemical reactions in solids//Russian chemical, Review 53 (11) (1984) 1025–1038, https://doi.org/10.1070/RC1984v053n11ABEH003138.
[51] at al. Hannes Kulla, Warming up for mechanosynthesis – temperature development in ball mills during synthesis, Chem. Commun. 53 (10) (2017), https://doi.org/10.1039/C6CC08950J.
[52] O. Lapshin, O. Ivanova, Macrokinetics of mechanochemical synthesis in heterogeneous systems: mathematical model and evaluation of thermokinetic

constants, Mater. Today Commun. 28 (2021), 102671, https://doi.org/10.1016/j.mtcomm.2021.102671.

[53] O. Lapshin, O. Ivanova, Macrokinetic Mechanosynthesis Model Comprising Multidirectional Factors Characterizing the Effect of Mechanical Treatment on the Combustion of Activated Mixtures, Powder Technology PTEC, 2022, 117419, https://doi.org/10.1016/j.powtec.2022.117419.

[54] V.K. Smolyakov, O.V. Lapshin, V.V. Boldyrev, Macroscopic theory of mechanochemical synthesis in heterogeneous systems, Int. J. Self-Propag. High-Temp. Synth. 16 (1) (2007) 1–11, https://doi.org/10.3103/S1061386207010013.

[55] V.K. Smolyakov, O.V. Lapshin, V.V. Boldyrev, Mechanochemical synthesis of nanosize products in heterogeneous systems: macroscopic kinetics, Int. J. Self-Propag. High-Temp. Synth. 17 (1) (2008) 20–29, https://doi.org/10.3103/S1061386208010020.

[56] O.V. Lapshin, Modelling mechanical and structural transformatijns in a binary powder mixture, Theor. Found. Chem. Eng. 54 (2) (2020) 349–356, https://doi.org/10.1134/S0040579520010121.

[57] G.S. Khodakov, Physical and chemical mechanics of grinding solids, Colloid J. 5 (1998) 684–697.

[58] Concise Encyclopedia of Self-Propagating High-Temperature Synthesis History, Theory, Technology and Products, Elsevier, 2017, p. 466.

[59] L.K. Doraiswamy, M.M. Sharma, Heterogeneous Reactions: Analysis, Examples and Reactor Design, John Wiely, New York, 1984, p. 538.

[60] O. Lapshin, O. Shkoda, O. Ivanova, S. Zelepugin, Discrete one-stage mechanochemical synthesis of titanium-nitride in a high, -Energy Mill//Metals 11 (2021) 1743, https://doi.org/10.3390/met11111743.

[61] J. Ding, T. Tsuzuki, P.G. McCormick, R. Street, Ultrafine Co and Ni particles prepared by mechanochemical processing, J. Phys. Appl. Phys. 29 (9) (1996) 2365–2369, https://doi.org/10.1088/0022-3727/29/9/020.

[62] E.P. Naiden, V.A. Zhuravlev, V.I. Itin, O.G. Terekhova, A.A. Magaeva, YuF. Ivanov, Magnetic properties and powders produced by mechanochemical synthesis from salt solutions, Phys. Solid State 50 (5) (2008) 894–900, https://doi.org/10.1134/S1063783408050156.

[63] YuM. Yevdokimov, V.I. Salyanov, V.I. Popenko, S.G. Skuridin, A.G. Pershina, A. A. Magaeva, E.V. Shtykova, I.A. Dadinova, Superparamagnetic cobalt ferrite nanoparticles «blow up» the spatial ordering of double-stranded DNA molecules, Biothysics 60 (3) (2015) 341–347, https://doi.org/10.1134/S0006350915030057.